NASA SPACE APPS CHALLENGE 2026

Astro Sağlık

Space Station Health Mission

A 3D game where kids monitor an astronaut's health, decide what is wrong and act fast to keep the crew safe.

Team: AstroPulse

Member: Azra Kurt (solo team)

Challenge: Create Health Monitoring Software for Astronauts on Space Missions



Full Presentation

Meet the team



Team picture is an AI-generated illustration

Azra Kurt

Solo team · Team AstroPulse

Our name: “Astro” = space, “Pulse” = heartbeat. We keep every astronaut’s pulse healthy.

Our challenge



Create Health Monitoring Software for Astronauts on Space Missions



Monitor

Track the key health signals of an astronaut



Understand

Turn numbers into a clear decision



Teach

Make it fun, so students learn space health

Why it matters: 5 hazards of spaceflight

NASA's Human Research Program names five hazards of human spaceflight (RIDGE):



Radiation

Higher cancer & tissue risk



Isolation

Stress, mood, team conflict



Distance

No quick trip to a doctor



Gravity

Bone & muscle loss, fluid shift



Hostile closed space

Germs, sleep, noise

Astronauts must notice problems early and act on their own. Our game teaches exactly that: watch the data, decide, act.

7 health indicators we monitor



Heart & vessels

Pulse, blood pressure



Bone & muscle

Bone density



Immune system

Body temperature



Radiation safety

Radiation dose



Mood

Stress level



Sleep & energy

Hours of sleep



Water & food

Body water

Bars slowly drop every mission day (faster as days pass). Below 25 → red alarm. At 0 → medical evacuation.

Our solution: Collect → Assess → Act

1 Collect data

Daily health scan: heart rate, blood pressure, SpO_2, temperature, bone density, radiation dose, stress, sleep, hydration

2 Assess

Read the numbers and choose the riskiest area or the right medical decision

3 Act

Go to the right module: exercise, eat, sleep, grow plants, call family, or shelter from radiation



In-game health scan (game UI is in Turkish)

7 live health bars drop over time. If one reaches 0 → medical evacuation, mission over.

A space station with 7 health modules



Blocky astronaut in the 3D station

- **Medical Center** Health scan & diagnosis
- **Gym** Treadmill & ARED workout
- **Kitchen** Build a balanced meal
- **Greenhouse** Water the plants
- **Comms Room** Mood check & family call
- **Sleep Pod** Rest and recharge
- **Radiation Shelter** Hide from solar storms

Health alarms and solar storms



Fever alarm: a beam and arrow show where to go



Solar flare: run to the shelter in 20 seconds

- 7 random health events: fever, heart rhythm, blurred vision (SANS), stress, bone pain, dehydration, insomnia
- Solar flares on mission days 2, 4 and 6 – race to the Radiation Shelter

Make the right medical call



Decision question + “Real space” fact box

Example

“Fever 38.2 °C, tired, sore throat. What is the best action?”

- Right answer: +150 points
- Wrong answer: −20, correct one shown
- Then a real NASA fact explains why

Players learn from mistakes, not just from success.

Health Report Card & scoring



End of mission: score, stars, all 7 bars and a tip

Action	Points
Daily health scan	+60
Correct assessment	+80
Health event solved	+150
Shelter during solar flare	+150
All 4 daily tasks	+100
Mission complete	200 + 3 × avg. health

If a bar hits 0 the mission ends early, but the score is still saved.

Play together in class



Add up to 8 players and choose 3, 5 or 7 days

Take turns

Each player plays, then sees their own score

Podium

Final ranking + all-time records

First-person view

See through the astronaut’s eyes (V key)

Any device

Keyboard & mouse or touch joystick

Built on NASA science

Human Research Program

5 hazards (RIDGE) → our 7 health indicators and events

ISS exercise & ARED

~2 hours of exercise a day protects bone and muscle

SANS

Fluid shift → vision problems event

Immune changes

Dormant viruses can reactivate → fever event

Veggie experiment

Astronauts grow lettuce → greenhouse module

Solar particle events

Shelter + AstroRad vest tested on Artemis I

Every quiz ends with a “Real space” fact box, so players learn the real science behind each choice.

How we built it

Three.js 3D

HTML + JavaScript, one single file

Works offline

Library saved locally – no internet needed

PC & tablet

Touch joystick, buttons, fullscreen

Sound

Made in code with Web Audio

Tested

Desktop and tablet views, every module

Development log

Every change recorded step by step

Use of Artificial Intelligence

Where and how AI was used

- An AI coding assistant (Claude) helped write and test the game code (HTML, JavaScript, Three.js)
- The team chose the challenge, the game idea, the health topics and the gameplay, and reviewed and tested every feature
- The team logo and team picture were made with an AI image tool – each is marked “AI-generated image”
- No NASA logos or mission insignia are used anywhere



Example: our AI-generated team picture with its watermark

Data & resources

NASA resources

- Human Research Program – Hazards of Human Spaceflight (nasa.gov/hrp/hazards)
- ISS exercise & ARED device
- SANS (Spaceflight Associated Neuro-ocular Syndrome)
- Immune system & latent virus research
- Veggie plant growth experiment
- Artemis I radiation research (AstroRad vest)

Tools & other resources

- Three.js r149 (MIT license)
- Fredoka font – Google Fonts (Open Font License)
- Claude – AI coding assistant
- AI image tool – logo & team picture (marked)
- All sounds generated in code

Impact & what comes next

Impact

- Students learn space health by playing
- Builds a real skill: read data → decide → act
- Free, offline, works on school tablets
- Whole class can play and compare scores

Next steps

- English version of the game
- Real wearable data (smartwatch heart rate)
- Difficulty levels and a Mars mission mode
- Downloadable health report (PDF)

Thank you! Team AstroPulse · Azra Kurt